Towards sustainable agriculture: A blueprint for European KPI-based farm-level assessment

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ABSTRACT ORIGINAL

Key Performance Indicator (KPI) methodologies have emerged as potent tools for estimating sustainability performance in agriculture. With the proliferation of data availability, there is a growing opportunity to conduct sustainability assessments at the farm level and transition from traditional practice-based policies to performance-based policies. In this paper, we propose a blueprint for a Europe-wide KPI-based farm level sustainability performance assessment and monitoring system. By drawing insights from literature on farm-level sustainability assessment systems and tools, we synthesized eight guiding principles and a 5-step process framework for developing and operationalizing a European Agricultural KPI system built around the potential advantages of performance-based policies. The guiding principles include defining sustainability, involving stakeholders, ensuring applicability across sectors and countries, using normative benchmarking, and implementing decision support approaches. The proposed process framework can guide public and private stakeholders in Europe through the establishment of sustainability goals, modeling linkages between objectives and available data, KPI development, online user interface creation, and adaptive management. By leveraging advanced analytics and the integration of diverse agricultural data sources, our system aims to provide policymakers with the opportunity to instrumentalize performance-based financial support mechanisms to incentivize the adoption of sustainable agricultural practices among farmers. The essential components of a policy structure supporting the effective and ethical implementation of the proposed KPI system are discussed. © 2025 The Author(s)